Lakshmi Sri Divya Raviprolu

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Skills

Languages: C++, Java, Python, JavaScript, TypeScript, SQL

Computer Science Fundamentals: & Technologies: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, DAA, UI/ UX Designing, Time-Series Forecasting, Scikit-learn, Data Pre-processing, Feature Engineering, Prompt Engineering, NodeJS, ReactJS

Professional Experience

Infosys Springboard Virtual Internship 6.0 Aug 2025 - Oct 2025

Intern

- Developed an Azure-based demand forecasting and capacity optimization system to predict CPU, storage, and active user demand across multiple cloud regions.
- Achieved 84.7% forecasting accuracy using LSTM, XGBoost, and ARIMA models tailored to regional demand patterns.
- Built data preprocessing and feature engineering pipelines by integrating Azure telemetry data, holiday calendars, and external indicators.
- Designed a Capacity Risk Matrix to classify cloud regions into utilization risk zones using 30-day consumption forecasts and infrastructure limits.
- Developed an automated recommendation engine that generated proactive scaling alerts and resource optimization insights for cloud capacity planning.
- **Technologies:** Python, Machine Learning, LSTM, XGBoost, ARIMA, Azure, Pandas, NumPy, Scikit-learn

AI for Sustainability Virtual Internship (IBM SkillsBuild) Dec 2025 - Jan 2026

Intern

- Submitted the MDR (Multi Drug Resistant) Sentinel project as the final internship deliverable, applying AI concepts to support hospital outbreak investigation and infection tracing.
- Applied Design Thinking methodologies to refine platform workflows and improve user-centric solution design.
- Designed a conceptual Agentic AI workflow using IBM Granite models for autonomous contact-tracing assistance.
- Explored Retrieval-Augmented Generation (RAG) techniques for semantic querying of medical reports and exposure logs.
- Developed advanced prompt engineering skills to optimize interactions with large language models and AI assistants.
- **Technologies:** Generative AI, IBM Granite, Agentic AI, RAG, Prompt Engineering, Design Thinking

Education

Everwin Vidhyashram CBSE School, Chennai, Tamil Nadu, India Jun 2021 - Jun 2022
Secondary Education (Class X) ***Percentage: 80.6%***

Everwin Vidhyashram CBSE School, Chennai, Tamil Nadu, India Jun 2023 - Jun 2024
Senior Secondary Education (Class XII) ***Percentage: 81.8%***

R.M.D. Engineering College, Tamil Nadu, India Aug 2024 - Expected Jun 2028
B.E. in Computer Science and Engineering ***CGPA: 8.61 / 10.00***
Relevant Coursework: Object-Oriented Programming, DBMS, Discrete Mathematics, Data Structures and Algorithms, DAA, Operating Systems, Computer Networks, Machine Learning, Data Mining, and Full Stack Development

Project Work

- **Global Recycling and Environmental Engagement Network (GREEN): Biomedical Waste Segregation (2025):** Engineered a smart, local-first waste classification application integrated with physical Arduino hardware (moisture, IR, and metal sensors). Programmed segregation logic to categorize materials automatically into dry, wet, metal, and hazardous bins. Developed a gamified dashboard based on global Sustainable Development Goals (SDGs) to drive user compliance. React, PostgreSQL, Arduino, IoT
- **Azure Demand Forecaster (2025):** Developed an end-to-end cloud resource demand forecasting pipeline to predict server CPU and storage needs. Engineered time-series data pipelines and trained XGBoost, LSTM, and ARIMA models to achieve a capacity prediction accuracy of 84.7%. Integrated a capacities risk matrix and real-time dashboard to flag resource utilization alerts. Python, Machine Learning (LSTM/XGBoost), Flask, React
- **Multi Drug Resistant (MDR) Sentinel (2025):** Designed a digital-twin simulation system for hospitals using Bluetooth Low Energy (BLE) tracking to trace staff and outpatient pathways. Implemented real-time contact tracing algorithms and retroactive infection replay engines to identify contamination nodes and predict multi-drug resistant (MDR) outbreak clusters. Node.js, IoT, BLE Sensors

Awards and Certificates

- Selected among the top 29 teams out of 168 participating teams for the project **G.R.E.E.N. (Global Recycling and Environmental Engagement Network): Biomedical Waste Segregation**, and awarded a grant of **Rs.5,000** during the NSD'25 National Science Day Celebrations organized by R.M.D. Engineering College.
- Earned **Elite + Gold certification** in NPTEL Introduction to Internet of Things, achieving a score of **93%** and ranking in the **top 1%** of participants nationwide.
- Achieved an Elite Certification with a final score of **72%** in Programming in Java, NPTEL, IIT Kharagpur.